已知 ， $x=x_0$ 时

$$\int_{-\infty}^{\infty} dx\ K(xt;xt_0) = \sum_n e^{-\frac{i}{\hbar}E_n(t-t_0)}$$ 求证.

· 对谐振子， （取 $t_0=0$）

$$K(x,t;x_,0) =$$

$$\sqrt{\frac{m\omega}{i2\pi\hbar\sin\omega t}}\exp\left[\frac{i}{\hbar}\frac{m\omega}{2\sin\omega t}\left(2x^2\cos\omega t - 2x^2\right)\right]$$

$$= \sqrt{\frac{m\omega}{i2\pi\hbar\sin\omega t}}\exp\left[\frac{im\omega x^2}{\hbar}\left(\frac{\cos\omega t - 1}{\sin\omega t}\right)\right]$$

$$= \sqrt{\frac{m\omega}{i2\pi\hbar\sin\omega t}}\exp\left[\frac{im\omega x^2}{\hbar}\left(-tg\frac{\omega t}{2}\right)\right]$$

$$\int_{-\infty}^{\infty} dx \; K(xt\,;\,x t_0)$$

$$= \int_{-\infty}^{\infty} dx \sqrt{\frac{m\omega}{i\,2\pi\hbar \sin\omega t}} \exp\left[-\frac{i\,m\omega x^2}{\hbar} \tan\frac{\omega t}{2} \right]$$

$$= \sqrt{\frac{m\omega}{i\,2\pi\hbar \sin\omega t}} \sqrt{\frac{\pi\hbar}{i\,m\omega \tan\frac{\omega t}{2}}}$$

$$= \frac{1}{2i \sin\frac{\omega t}{2}} = \frac{1}{e^{i\frac{\omega t}{2}} - e^{-i\frac{\omega t}{2}}}$$

$$= \frac{e^{-i\frac{\omega t}{2}}}{1 - e^{-i\omega t}} = e^{-i\frac{\omega t}{2}} \sum_{n=0}^{\infty} e^{-i n\omega t}$$

$$= \sum_{n=0}^{\infty} e^{-i\left(n+\frac{1}{2}\right)\omega t} = \sum_{n=0}^{\infty} e^{-\frac{i}{\hbar}\left(n+\frac{1}{2}\right)\hbar\omega t}$$

$$\Rightarrow \quad E_n = \left(n + \frac{1}{2}\right)\hbar\omega$$

已知，当 $x=x_0$ 时，有

$$K(xt;xt_0)=\sum_n e^{-\frac{i}{\hbar}E_n(t-t_0)}|\psi_n^{(0)}|^2 .$$

· 对谐振子 (以下取 $t_0=0$)

$$K(xt;x0)=\sqrt{\frac{m\omega}{\pi\hbar 2i\sin\omega t}}\exp\left\{-\frac{im\omega x^2}{\hbar}+g\frac{\omega t}{2}\right\}$$

$$=\sqrt{\frac{m\omega}{\pi\hbar}}\sqrt{\frac{1}{2i\sin\omega t}}\exp\left\{-\frac{m\omega x^2}{\hbar}\frac{1-e^{-i\omega t}}{1+e^{-i\omega t}}\right\}$$

$$=\sqrt{\frac{m\omega}{\pi\hbar}}\sqrt{\frac{1}{e^{i\omega t}-e^{-i\omega t}}}\exp\left\{-\frac{m\omega x^2}{\hbar}\frac{1-e^{-i\omega t}}{1+e^{-i\omega t}}\right\}$$

$$=\sqrt{\frac{m\omega}{\pi\hbar}}\sqrt{\frac{e^{-i\omega t}}{1-e^{-i2\omega t}}}\exp\left\{-\frac{m\omega x^2}{\hbar}\frac{1-e^{-i\omega t}}{1+e^{-i\omega t}}\right\}$$

$$= \sqrt{\frac{m\omega}{\pi\hbar}}\, e^{-\frac{i\omega t}{2}} \sqrt{\sum_{n=0} e^{-i2n\omega t}}$$

$$\times \exp\left[-\frac{m\omega x^2}{\hbar}\left(1-e^{-i\omega t}\right)\sum_{n=0}\left(-1\right)^n e^{-in\omega t}\right]$$

$$= \sqrt{\frac{m\omega}{\pi\hbar}}\, e^{-\frac{i\omega t}{2}}\left(1+e^{-i2\omega t}+e^{-i4\omega t}+\cdots\right)^{\frac{1}{2}}$$

$$\times \exp\left[-\frac{m\omega x^2}{\hbar}\left(1-e^{-i\omega t}\right)\left(1-e^{-i\omega t}+e^{-i2\omega t}-\cdots\right)\right]$$

$$= \sqrt{\frac{m\omega}{\pi\hbar}}\left(e^{-i\frac{\omega t}{2}}+\frac{2m\omega}{\hbar}x^2 e^{-i\frac{3\omega t}{2}}+\cdots\right) e^{-\frac{m\omega}{\hbar}x^2}$$

$$= \sqrt{\frac{m\omega}{\pi\hbar}}\left(e^{-\frac{i}{\hbar}\frac{\hbar\omega}{2}}+\frac{2m\omega}{\hbar}x^2 e^{-\frac{i}{\hbar}(1+\frac{1}{2})\hbar\omega t}+\cdots\right) e^{-\frac{m\omega}{\hbar}x^2}$$

$$= e^{-\frac{i}{\hbar}E_0 t}|\psi_0(x)|^2 + e^{-\frac{i}{\hbar}E_1 t}|\psi_1(x)|^2+\cdots$$

$$\Rightarrow \quad |\psi_0(x)|^2 = \sqrt{\frac{m\omega}{\pi\hbar}}\, e^{-\frac{m\omega x^2}{\hbar}},$$

得, $$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega x^2}{2\hbar}}$$

$$\Rightarrow \quad |\psi_1(x)|^2 = \sqrt{\frac{m\omega}{\pi\hbar}}\, 2m\omega x^2\, e^{-\frac{m\omega}{\hbar}x^2}$$

$$\text{得} \quad \psi_1(x) = \sqrt{\frac{2m\omega}{\hbar}}\, x\, \psi_0(x) \quad .$$

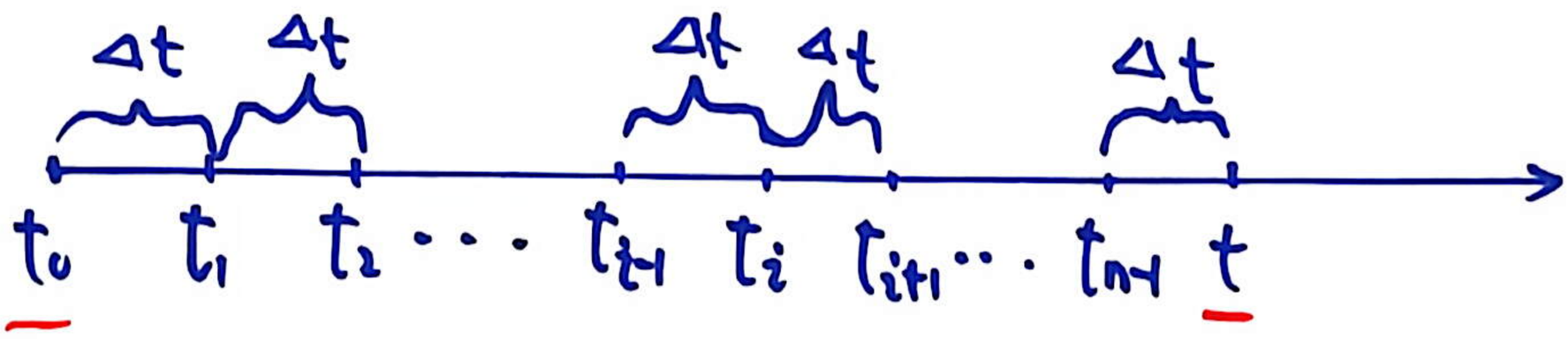
4.4 传播子的计算(薛-路径积分)

(a) 相空间中的路径积分

$\hat{H}(\hat{x},\hat{p},t)$ —— 哈密顿算符 (Weyl 正规化)

- $K(xt;x_0t_0) = \langle x|\hat{U}(t,t_0)|x_0\rangle$

- 将 $(t-t_0)$ 分成 N 个等分



$$t_{i+1}-t_i = \Delta t = \frac{t-t_0}{N} \; ; N\to\infty \;,\; \Delta t \to 0$$

利用传播子的组合规则，可得 (取 $N\to\infty$)

$$K(xt;x_0t_0)=\lim_{N\to\infty}\int\left(\prod_{j=1}^{N-1}dx_j\right)\prod_{i=0}^{N-1}K(x_{i+1}t_{i+1};x_it_i)$$

其中. $\begin{cases}x_N=x\\ t_N=t\end{cases}$

一般式 $K(x_{i+1}t_{i+1};x_it_i)$ 的计算

$$K(x_{i+1}t_{i+1};x_it_i)=\langle x_{i+1}|\hat{U}(t_{i+1},t_i)|x_i\rangle,$$

其中 $t_{i+1}-t_i=\Delta t\to 0$，(当 $N\to\infty$ 时)

$\hat{H}$ 可以近似为与时间无关.

$$\Rightarrow\quad \hat{U}(t_{i+1},t_i)=e^{-\frac{i}{\hbar}\hat{H}(\hat{x},\hat{p},t)(t_{i+1}-t_i)}$$

$$\approx 1-\frac{i}{\hbar}\hat{H}(\hat{x},\hat{p},t)(t_{i+1}-t_i)$$

$$K(x_{i+1}t_{i+1};x_i t_i)$$

$$\approx \langle x_{i+1}| \left[1 - \frac{i}{\hbar}\hat{H}(\hat{x},\hat{p},t)(t_{i+1}-t_i)\right]|x_i\rangle$$

$$= \int dp_i \langle x_{i+1}|p_i\rangle\langle p_i|\left[1 - \frac{i}{\hbar}\hat{H}(\hat{x},\hat{p},t)(T_{i+1}-t_i)\right]|x_i\rangle$$

其中. $\langle p_i|\hat{H}(\hat{x},\hat{p},t)|x_i\rangle = H(x_i,p_i,t)\langle p_i|x_i\rangle$

$$= \int dp_i \langle x_{i+1}|p_i\rangle\langle p_i|\left[1 - \frac{i}{\hbar}H(x_i,p_i,t_i)(t_{i+1}-t_i)\right]|x_i\rangle$$

$$\approx \int dp_i \langle x_{i+1}|p_i\rangle\langle p_i|x_i\rangle e^{-\frac{i}{\hbar}H(x_i,p_i,t_i)(t_{i+1}-t_i)}$$

$$= \int \frac{dp_i}{2\pi\hbar} e^{\frac{i}{\hbar}p_i(x_{i+1}-x_i)} e^{-\frac{i}{\hbar}H(x_i,p_i,t)(t_{i+1}-t_i)}$$

$$= \int \frac{dp_i}{2\pi\hbar} e^{\frac{i}{\hbar}(t_{i+1}-t_i)\left[p_i\frac{(x_{i+1}-x_i)}{(t_{i+1}-t_i)} - H(x_i,p_i,t_i)\right]}$$

$\Rightarrow$

$$K(xt;x_0t_0) =$$

$$\lim_{N\to\infty} \int \left(\prod_{j=1}^{N-1} dx_j\right) \int \left(\prod_{i=0}^{N-1} \frac{dp_i}{2\pi\hbar}\right)$$

$$\times \exp\left[\frac{i}{\hbar}\Delta t \sum_{i=0}^{N-1} \left(p_i \frac{(x_{i+1}-x_i)}{\Delta t} - H(x_i p_i t_i)\right)\right]$$

$$\lim_{N\to\infty} \begin{cases} \int \left(\prod_{j=1}^{N-1} dx_i\right) \to \int D[x(t)] \\ \int \left(\prod_{i=0}^{N-1} \frac{dp_i}{2\pi\hbar}\right) \to \int D[p(t)] \\ \Delta t \sum_{i=0}^{N-1} f(t_i) \to \int_{t_0}^{t} d\tau\, f(\tau) \\ \frac{x_{i+1}-x_i}{\Delta t} \to \dot{x}(t_i) \end{cases}$$

$$K(xt;x_0t_0) = \int D[x(\tau)] \int D[p(\tau)]$$

$$\times \exp\left[\frac{i}{\hbar} \int_{t_0}^{t} d\tau \left[p(\tau)\dot{x}(\tau) - H(x(\tau), p(\tau), \tau)\right]\right]$$

(b) 位形空间中的路径积分

$$\hat{H}(\hat{x},\hat{p},t) = \frac{\hat{p}^2}{2m} + \hat{V}(\hat{x},t)$$

- $K(x_{i+1}\,t_{i+1};\,x_i\,t_i)$

$$= \int \frac{dp_i}{2\pi\hbar} e^{\frac{i}{\hbar}(t_{i+1}-t_i)\left[p_i \frac{(x_{i+1}-x_i)}{(t_{i+1}-t_i)} - H(x_i,p_i\,t_i)\right]}$$

代入 $H(x_i,p_i\,t_i) = \frac{p_i^2}{2m} + V(x_i\,t_i)$

$$K(x_{i+1}\,t_{i+1};\,x_i\,t_i)$$

$$= \int \frac{dp_i}{2\pi\hbar} e^{\frac{i}{\hbar}(t_{i+1}-t_i)\left[p_i \frac{(x_{i+1}-x_i)}{(t_{i+1}-t_i)} - \frac{p_i^2}{2m} - V(x_i\,t_i)\right]}$$

$$= \int \frac{dp_i}{2\pi\hbar} e^{\frac{i}{\hbar}\left[p_i(x_{i+1}-x_i) - \frac{p_i^2}{2m}(t_{i+1}-t_i)\right]} \times e^{-\frac{i}{\hbar}(t_{i+1}-t_i)V(x_i\,t_i)}$$

$$= \sqrt{\frac{m}{i2\pi\hbar\Delta t}}\, e^{\frac{i}{\hbar}\frac{m}{2\Delta t}(x_{i+1}-x_i)^2} \times e^{-\frac{i}{\hbar}\Delta t\,V(x_i\,t_i)}$$

这里，利用了Fresnel积分公式

さp

$$K(x_{i+1}t_{i+1}, x_i t_i) =$$

$$\sqrt{\frac{m}{i2\pi\hbar\Delta t}} \exp\left[\frac{i}{\hbar}\Delta t\left\{\frac{m}{2}\left(\frac{x_{i+1}-x_i}{\Delta t}\right)^2 - V(x_i t_i)\right\}\right]$$

$\Rightarrow$

$$K(xt;x_0t_0) = \lim_{N\to\infty}\left(\frac{m}{i2\pi\hbar\Delta t}\right)^{\frac{N}{2}}$$

$$\int_{-\infty}^{\infty}\left(\prod_{j=1}^{N-1}dx_j\right)\prod_{i=0}^{N-1}\exp\left[\frac{i}{\hbar}\Delta t\left\{\frac{m}{2}\left(\frac{x_{i+1}-x_i}{\Delta t}\right)^2 - V(x_i t_i)\right\}\right]$$

$$K(xt;x_0t_0) = \lim_{N\to\infty}\left(\frac{m}{i2\pi\hbar\Delta t}\right)^{\frac{N}{2}}$$

$$\int_{-\infty}^{\infty}\left(\prod_{j=1}^{N-1}dx_j\right)\exp\left[\frac{i}{\hbar}\Delta t\sum_{i=0}^{N-1}\left\{\frac{m}{2}\left(\frac{x_{i+1}-x_i}{\Delta t}\right)^2 - V(x_i,t_i)\right\}\right]$$

路径积分？

$$\int \left(\prod_{i=1}^{N-1} dx_i\right)\left(\frac{m}{i2\pi\hbar\Delta t}\right)^{\frac{N}{2}} \to \int D[x(t)]$$

$$\lim_{N\to\infty}\begin{cases} \dfrac{(x_{j+1}-x_j)}{\Delta t} \to \dot{x}(t_j) \\ \Delta t \sum_{j=0}^{N-1} f(t_j) \to \int_{t_0}^{t} d\tau\, f(\tau) \end{cases}$$

$$K(xt;x_0t_0) = \int D[x(t)]\ \exp\left[\frac{i}{\hbar}\int_{t_0}^{t} d\tau \left(\frac{m}{2}\dot{x}^2(\tau) - V(x(\tau),\tau)\right)\right]$$

$$= \int D[x(t)]\ \exp\left[\frac{i}{\hbar}\int_{t_0}^{t} d\tau\, L(\dot{x}(\tau), x(\tau), \tau)\right]$$

$$= \int D[x(t)]\ \exp\left[\frac{i}{\hbar} S(\dot{x}(t), x(t))\right]$$

其中. $L = \frac{m}{2}\dot{x}^2 - V$ （拉氏量），

$S = \int dt\, L$ （作用量）.

例：自由粒子系统

$$\hat{H} = \frac{\hat{p}^2}{2m}$$

$$K(xt;x_0t_0) = \lim_{N\to\infty} \left(\frac{m}{i2\pi\hbar\Delta t}\right)^{\frac{N}{2}} \int dx_1\, dx_2 \cdots dx_{N-1}$$

$$\exp\left[\frac{i}{2\hbar\Delta t}\frac{m}{}\sum_{i=0}^{N-1}(x_{i+1}-x_i)^2\right]$$

$$= \lim_{N\to\infty}\left(\frac{m}{i2\pi\hbar\Delta t}\right)^{\frac{N}{2}} \int dx_1\, dx_2 \cdots dx_{N-1}$$

$$\exp\left[i\frac{m}{2\hbar\Delta t}\left\{(x_1-x_0)^2+(x_2-x_1)^2+\cdots(x_N-x_{N-1})^2\right\}\right]$$

利用公式：

$$\int dx_1\, dx_2 \cdots dx_{N-1} \exp\left\{i\alpha\left[(x_1-x_0)^2+(x_2-x_1)^2+\cdots(x_N-x_{N-1})^2\right]\right\}$$

$$= \left(\frac{i\pi}{\alpha}\right)^{\frac{N-1}{2}} \frac{1}{\sqrt{N}} \exp\left[\frac{i\alpha}{N}(x_N-x_0)^2\right]$$

代入 $\alpha = \frac{m}{2\pi\hbar\Delta t}$ ，得

$$K(x_1; x_0 t_0) =$$

$$\lim_{N\to\infty} \left(\frac{m}{i2\pi\hbar\Delta t}\right)^{\frac{N}{2}} \cdot \left(\frac{i\pi}{m/2\hbar\Delta t}\right)^{\frac{N-1}{2}} \frac{1}{\sqrt{N}}$$

$$\exp\left[\frac{im}{N2\hbar\Delta t}(x-x_0)^2\right]$$

$$= \lim_{N\to\infty} \left(\frac{i\pi}{m/2\hbar\Delta t}\right)^{-\frac{1}{2}} \frac{1}{\sqrt{N}} \exp\left[\frac{im}{N2\hbar\Delta t}(x-x_0)^2\right]$$

$$= \left(\frac{m}{i2\pi\hbar N\Delta t}\right)^{\frac{1}{2}} \exp\left[\frac{im}{2\hbar N\Delta t}(x-x_0)^2\right]$$

$$= \left(\frac{m}{i2\pi\hbar(t-t_0)}\right)^{\frac{1}{2}} \exp\left[\frac{im}{2\hbar(t-t_0)}(x-x_0)^2\right].$$

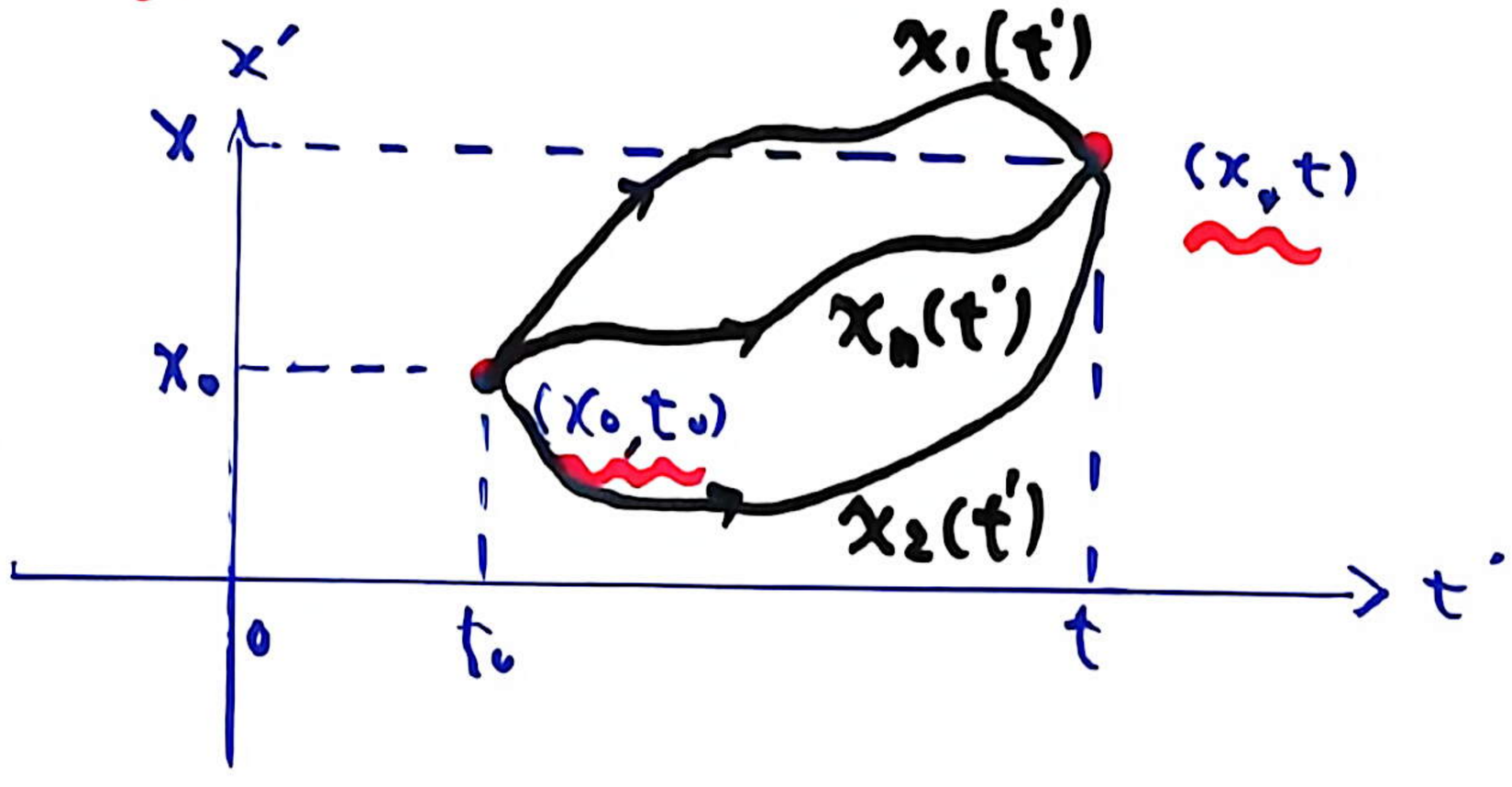
ok!

§2.5 Feynman 路径积分

· Feynman 传播子 : $K_F(xt;x_0t_0)$

定义式 : $$\psi(x,t)=\int_{-\infty}^{\infty}dx_0\,K_F(xt;x_0t_0)\,\psi(x_0,t_0)$$

$K_F(xt;x_0t_0)$ 的计算 ⇒



1. 画出连接 (x_0,t_0) 和 (x,t) 点的全部路径：

$x_1(t')$, $x_2(t')$ … $x_n(t')$ …

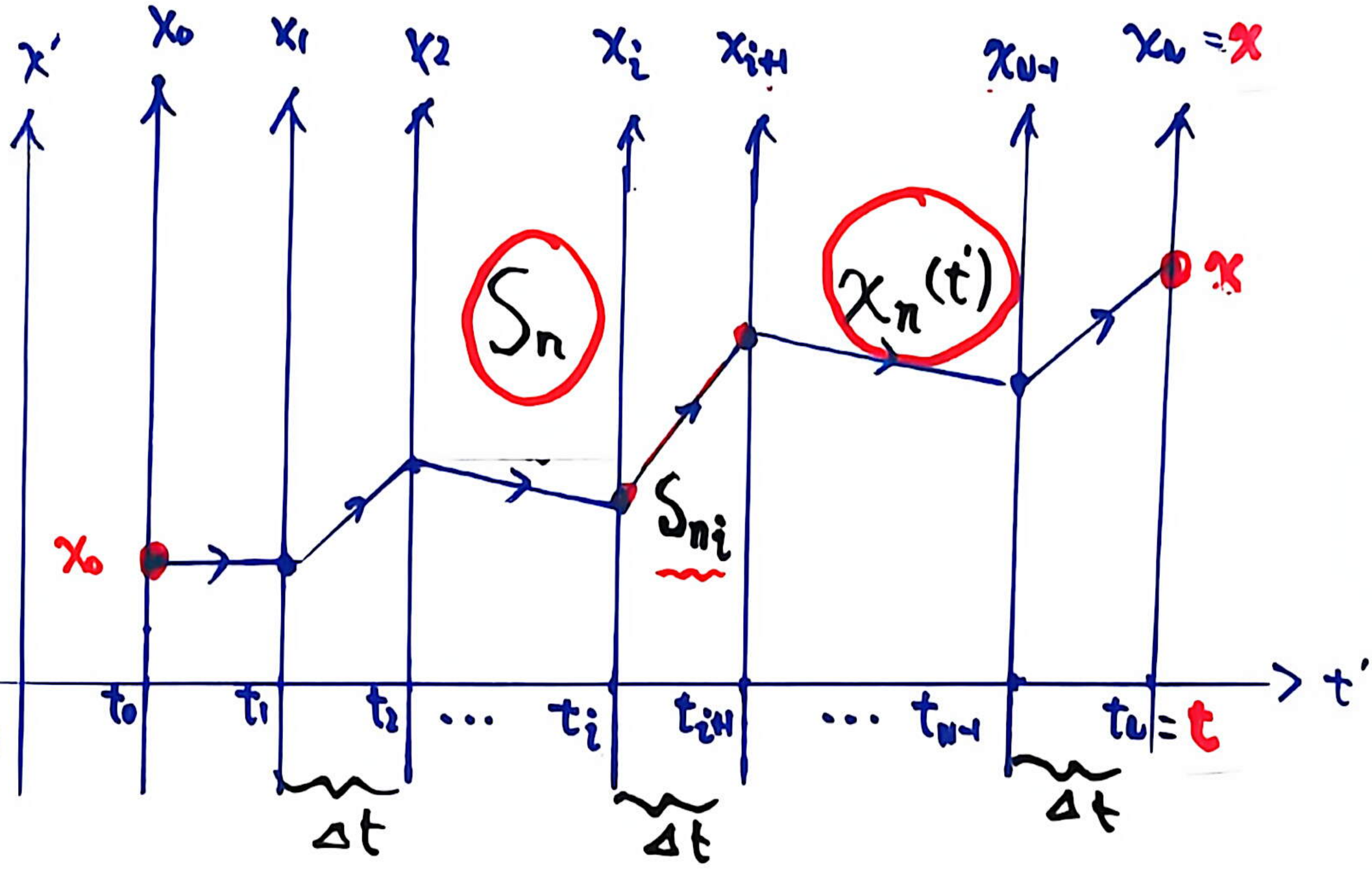
2. 对第 n 条路径 $x_n(x')$ 算出作用量

$$S_n=\int_{t_0}^{t}L_n\left(x_n(t'),\dot{x}_n(t'),t'\right)dt'$$

经典拉氏量. （泛函）

3. $K_F(xt;x_0t_0) \propto \sum_n \exp\left[\frac{i}{\hbar} S_n\right]$

全部路径



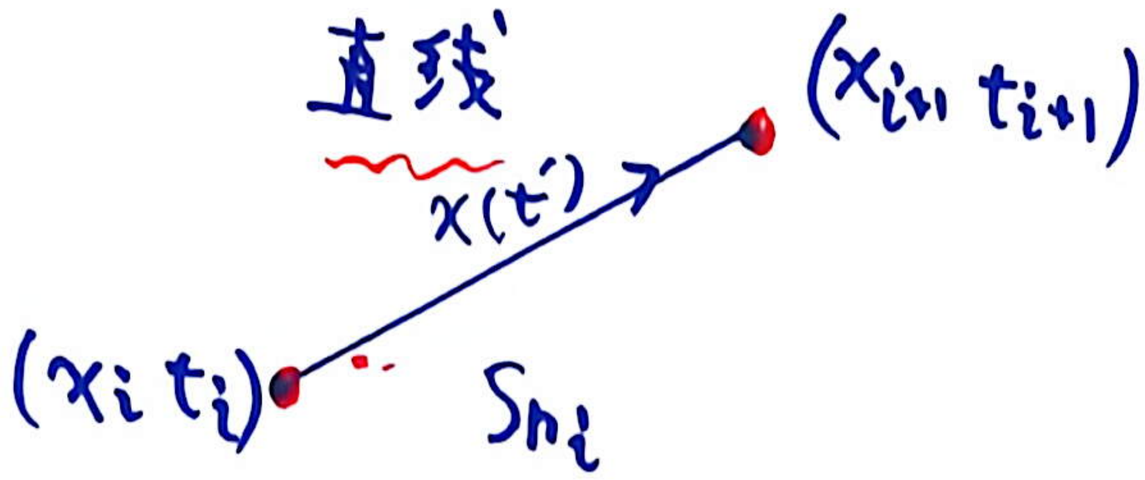
$$\left(\frac{t - t_0}{N} = \Delta t \right)$$

$N \to \infty$

$$S_n = \sum_{i=0}^{N-1} S_{ni}$$

$$S_{ni} \equiv \int_{t_i}^{t_{i+1}} L_{ni}\left(x_{ni}(t'), \dot{x}_{ni}(t'), t' \right) dt'$$

자유입자 $L(\dot{x}) \equiv \frac{m}{2}\dot{x}^2(t) - V(x,t)$



- $x(t') = x_i + \dfrac{(x_{i+1} - x_i)}{t_{i+1} - t_i}(t' - t_i)$.

直线方程

(=点式)

$$\Rightarrow \quad \dot{x}(t') = \frac{x_{i+1} - x_i}{t_{i+1} - t_i}$$

- $L \equiv \frac{m}{2}\dot{x}^2(t') - V(x(t'), t')$

$$\Rightarrow \quad L_{ni}^{(t')} = \frac{m}{2}\left[\frac{x_{i+1} - x_i}{t_{i+1} - t_i}\right]^2 - V[x_i, t']$$

- $$S_{ni} = \int_{t_i}^{t_{i+1}} \left[\frac{m}{2} \left(\frac{x_{i+1} - x_i}{t_{i+1} - t_i} \right)^2 - V(x_i, t') \right] dt'$$

$$= \Delta t \left[\frac{m}{2} \left(\frac{x_{i+1} - x_i}{t_{i+1} - t_i} \right)^2 - V(x_i\ t_i) \right] .$$

- $$S_n = \sum_{i=0}^{N-1} S_{ni}$$

$$= \sum_{i=0}^{N-1} \Delta t \left[\frac{m}{2} \left(\frac{x_{i+1} - x_i}{t_{i+1} - t_i} \right)^2 - V(x_i, t_i) \right]$$

- $$K_F(xt; x_0 t_0) \propto \sum_{\text{全ての経路}} \exp\left[\frac{i}{\hbar} S_n \right]$$

- $$K_F(xt; x_0 t_0) =$$

$$\lim_{N \to \infty} \int_{-\infty}^{\infty} \left(\prod_{j=1}^{N-1} dx_j \right) A \exp\left[\frac{i}{\hbar} \Delta t \sum_{i=0}^{N-1} \left(\frac{m}{2} \left(\frac{x_{i+1} - x_i}{t_{i+1} - t_i} \right)^2 - V(x_i, t_i) \right) \right] .$$

★ Feynman 传播子的初始条件及传递性.

- $$\psi(x,t) \equiv \int_{-\infty}^{\infty} dx'\ K_F(xt;x't_0)\,\psi(x',t_0)$$

上式中，令 $t=t_0$，则有

$$\psi(x,t_0) = \int_{-\infty}^{\infty} dx'\ K_F(xt_0;x't_0)\,\psi(x',t_0)$$

$$\Rightarrow \quad K_F(xt_0;x't_0) = \delta(x-x') \qquad \text{(初始条件)}$$

- $$\psi(x,t) = \int_{-\infty}^{\infty} dx_0\ K_F(xt;x_0t_0)\,\psi(x_0,t_0) \qquad (t>t_1>t_0)$$

又
$$\begin{cases} \psi(xt) = \int_{-\infty}^{\infty} dx_1\ K_F(xt;x_1t_1)\,\psi(x_1,t_1) \\ \psi(x_1t_1) = \int_{-\infty}^{\infty} dx_0\ K_F(x_1t_1;x_0t_0)\,\psi(x_0t_0) \end{cases}$$

比较

$$\Rightarrow \quad \psi(xt) = \int_{-\infty}^{\infty} dx_1 \int_{-\infty}^{\infty} dx_0\ K_F(xt;x_1t_1)\,K_F(x_1t_1;x_0t_0)\,\psi(x_0t_0)$$

$$\Rightarrow \quad K_F(xt;x_0t_0) = \int_{-\infty}^{\infty} dx_1\, K_F(xt;x_1t_1)\,K_F(x_1t_1;x_0t_0) \qquad \text{(传递性)}$$

$\Rightarrow$

$$K_F(x_t t; x_0 t_0) = \int_{-\infty}^{\infty} \left(\prod_{j=1}^{N-1} dx_j \right) \prod_{i=0}^{N-1} K_F(x_{i+1} t_{i+1}; x_i t_i) \quad ,$$

其中，$K_F(x_{i+1} t_{i+1}; x_i t_i) \equiv$

$$A^{\frac{1}{2}} \exp\left[\frac{i}{\hbar} \Delta t \left(\frac{m}{2} \left(\frac{x_{i+1} - x_i}{\Delta t} \right)^2 - V(x_i, t_i) \right) \right] .$$

Δt 为小量 $\Rightarrow$ 短时间传播子

4 Feynman 路径积分与 Schrödinger 方程

- $$\psi(x,t) = \int_{-\infty}^{\infty} dx' \, K_F(xt;x't_0)\,\psi(x't_0) \quad .$$

$$t_0 \to t \ , \quad t \to t+\epsilon \qquad (\epsilon \text{为小量})$$

$$\Rightarrow \ \psi(x,t+\epsilon) = \int_{-\infty}^{\infty} dx' \, K_F(x,t+\epsilon;x't)\,\psi(x't) \quad .$$

其中. $K_F(x,t+\epsilon;x't) =$ _____

$$\underbrace{A^{-1}}_{\sim\sim} \exp\left[\frac{i}{\hbar}\epsilon\left(\frac{m}{2}\left(\frac{x-x'}{\epsilon}\right)^2 - V(x',t)\right)\right] \quad .$$

- 令 $x' = x+\xi$, $dx' = d\xi$

$$\Rightarrow \ \psi(x,t+\epsilon) = A^{-1}\int_{-\infty}^{\infty} d\xi \ \exp\left[\frac{im\xi^2}{2\hbar\epsilon}\right]$$

$$\times \ \exp\left[-\frac{i\epsilon}{\hbar}V(x+\xi,t)\right]\psi(x+\xi,t) \quad .$$

· 积分中实际有贡献的区域：

$$\frac{m\xi^2}{2\hbar\epsilon} \lesssim 1 . \quad \epsilon 小 \Rightarrow \xi 小，相积主贡献$$

$$\Rightarrow \psi(x+\xi,t) \approx \psi(x,t) + \xi\frac{\partial\psi(x,t)}{\partial x} + \frac{1}{2}\xi^2\frac{\partial^2\psi(x,t)}{\partial x^2} .$$

$$K \epsilon V(x+\xi,t) \approx \epsilon V(x,t) .$$

$$\Rightarrow \exp\left[-\frac{i\epsilon}{\hbar}V(x+\xi,t)\right] \approx \exp\left[-\frac{i\epsilon}{\hbar}V(x,t)\right]$$

$$\approx 1 - \frac{i\epsilon}{\hbar}V(x,t) .$$

$$K 等式左边： \quad \psi(x,t+\epsilon) \approx \psi(x,t) + \epsilon\frac{\partial\psi(x,t)}{\partial t} .$$

$$\Rightarrow$$

$$\psi(x,t) + \epsilon\frac{\partial\psi(x,t)}{\partial t} = A^{N}\int_{-\infty}^{\infty} d\xi \exp\left[\frac{im\xi^2}{2\hbar\epsilon}\right]$$

$$\times\left[1 - \frac{i}{\hbar}\epsilon V(x,t) + \xi\frac{\partial}{\partial x} + \frac{1}{2}\xi^2\frac{\partial^2}{\partial x^2}\right]\psi(x,t) .$$

积分：

$$\int_{-\infty}^{\infty} d\xi \exp\left[\frac{im}{2\hbar\epsilon}\xi^2\right] = \sqrt{\frac{2\pi i\hbar\epsilon}{m}} .$$

$$\int_{-\infty}^{\infty} d\xi \exp\left[\frac{im}{2\hbar\epsilon}\xi^2\right]\xi = 0 , \quad (\text{奇函数})$$

$$\int_{-\infty}^{\infty} d\xi \exp\left[\frac{im}{2\hbar\epsilon}\xi^2\right]\xi^2 = \frac{i\hbar\epsilon}{m}\sqrt{\frac{2\pi i\hbar\epsilon}{m}} .$$

$$\Rightarrow$$

$$\psi(x,t) + \epsilon\frac{\partial\psi(x,t)}{\partial t} =$$

$$A^{N+1}\sqrt{\frac{2\pi i\hbar\epsilon}{m}}\left[1 - \frac{i\epsilon}{\hbar}V(x,t)\right]\psi(x,t)$$

$$+ A^{N+1}\frac{1}{2}\frac{i\hbar\epsilon}{m}\sqrt{\frac{2\pi i\hbar\epsilon}{m}}\frac{\partial^2\psi(x,t)}{\partial x^2} .$$

· 忽略所有高阶无穷小量，并要求上等式成立

$$\Rightarrow \quad \psi(x,t+\varepsilon) = A^{\frac{1}{2}}\sqrt{\frac{2\pi i\hbar\varepsilon}{m}}\ \psi(x,t)$$

$$\Rightarrow \quad A^{\frac{1}{2}}\sqrt{\frac{2\pi i\hbar\varepsilon}{m}} = 1 \quad , \quad \boxed{A^{\frac{1}{2}} = \sqrt{\frac{m}{2\pi i\hbar\varepsilon}}}.$$

将 $A^{\frac{1}{2}} = \sqrt{\frac{m}{2\pi i\hbar\varepsilon}}$ 代入 ① 式中，得

$$\varepsilon\ \frac{\partial\psi(x,t)}{\partial t} = -\frac{i\varepsilon}{\hbar}V(x,t)\psi(x,t) + \frac{i\hbar}{2m}\varepsilon\frac{\partial^2\psi(x,t)}{\partial x^2}$$

消去 ε，得

$$i\hbar\frac{\partial\psi(x,t)}{\partial t} = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x,t)\right]\psi(x,t).$$

$$\Rightarrow \quad i\hbar\frac{\partial}{\partial t}K_F(xt;x_0t_0) = \hat{H}\,K_F(xt;x_0t_0).$$

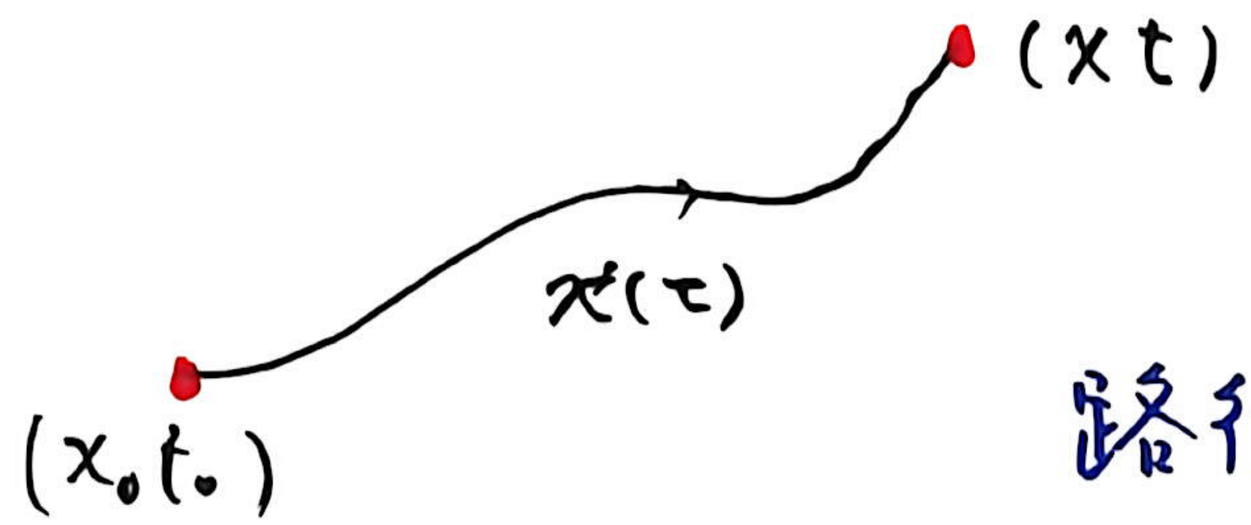
初始条件：$K_F(xt_0;x_0t_0)$

$$= \delta(x-x_0)$$

传播子的计算 （利用经典作用量）

· 作用量： $S = \int_{t_0}^{t} d\tau\, L(\dot{x}(\tau), x(\tau), \tau)$

拉氏量



路径的泛函.

例：

(1) 一维自由粒子系统

$$L = \frac{m}{2}\dot{x}^2$$

(2) 一维线性谐振子

$$L = \frac{m}{2}\dot{x}^2 - \frac{m}{2}\omega^2 x^2$$

· 经典作用量 $S_c(xt; x_0 t_0)$ （最小作用量原理）

$$\boxed{\frac{\delta S_c}{\delta x(t)} = 0} \Rightarrow \left(\frac{\partial L}{\partial x} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}}\right)_c = 0$$

拉氏方程

· 拉氏方程的导出

· $S = \int_{t_0}^{t} dt' \; L[\dot{x}(t'), x(t')]$

· $\frac{\delta S}{\delta x(t)} = \int_{t_0}^{t} dt' \; \frac{\delta L[\dot{x}(t'), x(t')]}{\delta x(t)}$

· 变分的定义：

$$\frac{\delta F[x(t')]}{\delta x(t)} = \lim_{\epsilon \to 0} \frac{F[x(t') + \epsilon \delta(t'-t)] - F[x(t')]}{\epsilon}$$

$$\frac{\delta S}{\delta x(t)} = 0 \quad \Rightarrow \quad \frac{\partial L\left[\dot{x}(t), x(t)\right]}{\partial x(t)} - \frac{d}{dt}\frac{\partial L\left[\dot{x}(t), x(t)\right]}{\partial \dot{x}(t)} = 0$$

קיבלנו

$$= -m\ddot{x}(\tau) - \frac{d}{dx(t)} V\left[x(t)\right].$$

$$= \int_{t_1}^{t_2} dt' \left[m\dot{x}(\tau')\frac{d}{dt'}\delta(t'-\tau) - \frac{d}{dx(\tau')} V\left[x(\tau')\right] \delta(t'-\tau) \right]$$

$$\Rightarrow \frac{\delta S}{\delta x(t)} = \frac{\partial L\left[\dot{x}(t), x(t)\right]}{\partial x(t)} - \frac{d}{dt}\frac{\partial L\left[\dot{x}(t), x(t)\right]}{\partial \dot{x}(t)} \cdot$$

$$= -m\ddot{x}(\tau) - \frac{d}{dx(\tau)} V\left[x(\tau)\right].$$

$$\Rightarrow \frac{\delta S}{\delta x(t)} = \frac{\partial L\left[\dot{x}(t), x(t)\right]}{\partial x(t)} - \frac{d}{dt}\frac{\partial L\left[\dot{x}(t), x(t)\right]}{\partial \dot{x}(t)} = 0$$

$$L\left(\dot{x}(t), x(t)\right) = \frac{m}{2}\dot{x}^2(t) - V\left(x(t)\right)$$

例1.1：自由粒子系统的经典作用量

- 拉氏量：$L = T - V = T = \frac{m}{2}\dot{x}^2$

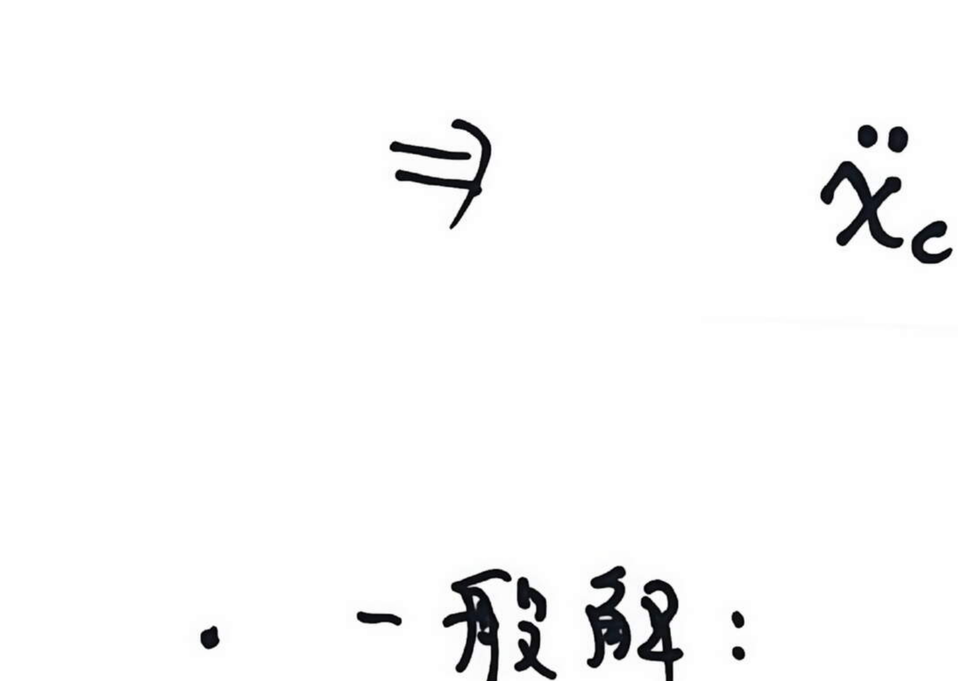
动能 势能

- 拉氏方程：$\frac{\partial L}{\partial x} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = 0$

$\frac{\partial L}{\partial x} = 0$ ； $\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = m\ddot{x}$

$\Rightarrow$ $\ddot{x}_c = 0$ （牛顿运动方程）

- 一般解：$x_c(t) = x_0 + v_0 t$ （直线）



边界条件：

$$\begin{cases} x_c(t_1) = x_1 = x_0 + v_0 t_1 \\ x_c(t_2) = x_2 = x_0 + v_0 t_2 \end{cases}$$

解出：

$$\begin{cases} x_0 = x_1 - \dfrac{x_1 - x_2}{t_1 - t_2} t_1 \\ v_0 = \dfrac{x_1 - x_2}{t_1 - t_2} \end{cases}$$

$$\Rightarrow \boxed{x_c(t) = x_1 + \left(\frac{x_1 - x_2}{t_1 - t_2}\right)(t - t_1)} \quad (\text{两点式})$$

$$\underline{\dot{x}_c(t) = \frac{x_1 - x_2}{t_1 - t_2}} \quad (\text{与 } t \text{ 无关})$$

· 经典拉氏量：$\underline{L_c = \frac{m}{2}\left(\frac{x_1 - x_2}{t_1 - t_2}\right)^2}$

· 经典作用量：$S_c = \int_{t_1}^{t_2} dt \; L_c$

$$= \frac{m}{2}\left(\frac{x_1 - x_2}{t_1 - t_2}\right)^2 (t_2 - t_1),$$

即

$$\boxed{S_c = \frac{m}{2}\frac{(x_2 - x_1)^2}{t_2 - t_1}}$$

例2、一维线性谐振子体系的经典作用量.

· 拉氏量： $L = T - V = \frac{m\dot{x}^2}{2} - \frac{m\omega^2 x^2}{2}$

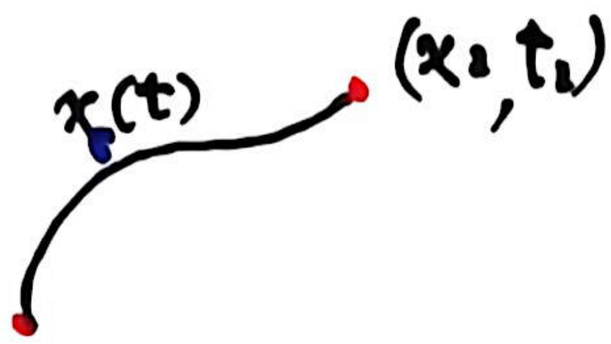
· 拉氏方程： $\frac{\partial L}{\partial x} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = 0$

$\frac{\partial L}{\partial x} = -m\omega^2 x$ ； $\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = m\ddot{x}$

$\Rightarrow$ $\ddot{x}_c + \omega^2 x_c = 0$ （运动方程）

· 通解： $x(t) = A\cos\omega t + B\sin\omega t$

边界条件：



$$\begin{cases} x_c(t_1) = A\cos\omega t_1 + B\sin\omega t_1 = x_1 \\ x_c(t_2) = A\cos\omega t_2 + B\sin\omega t_2 = x_2 \end{cases}$$

可解出：

$$A = \frac{x_1 \sin\omega t_2 - x_2 \sin\omega t_1}{\sin\omega(t_2 - t_1)}$$

$$B = -\frac{x_1 \cos\omega t_2 - x_2 \cos\omega t_1}{\sin\omega(t_2 - t_1)}$$

$$\Rightarrow \quad \boxed{x_c(t) = \frac{x_2 \sin\omega(t-t_1) - x_1 \sin\omega(t-t_2)}{\sin\omega(t_2 - t_1)}}$$

$$\boxed{\dot{x}_c(t) = \frac{\omega x_2 \cos\omega(t-t_1) - \omega x_1 \cos\omega(t-t_2)}{\sin\omega(t_2 - t_1)}}$$

$$S_c = \int_{t_1}^{t_2} dt\, L_c = \int_{t_1}^{t_2} dt \left[\frac{m}{2}\dot{x}_c^2(t) - \frac{m\omega^2}{2} x_c^2(t) \right]$$

$$= \frac{m}{2} \int_{t_1}^{t_2} dt \left(\dot{x}_c^2(t) - \omega^2 x_c^2(t) \right)$$

부-고전적분 :

$$\int_{t_1}^{t_2} dt\ \ddot{\chi}_c(t)\ \chi_c(t) = \chi_c\dot{\chi}_c\Big|_{t_1}^{t_2} - \int_{t_1}^{t_2} dt\ \ddot{\chi}_c(t)\ \chi_c(t)$$

$$\Rightarrow \quad S_c = \frac{m}{2}\chi_c\dot{\chi}_c\Big|_{t_1}^{t_2} - \frac{m}{2}\int_{t_1}^{t_2} dt\Big(\ddot{\chi}_c(t) + \omega^2\chi_c(t)\Big)\chi_c(t)$$

$$= \frac{m}{2}\ \chi_c(t)\,\dot{\chi}_c(t)\Big|_{t_1}^{t_2}$$

$$= \frac{m}{2}\Big[\ \chi_c(t_2)\,\dot{\chi}_c(t_2) - \chi_c(t_1)\,\dot{\chi}_c(t_1)\Big]$$

$$= \frac{m\omega}{2}\ \frac{\big[\ (\chi_1^2+\chi_2^2)\cos\omega(t_2-t_1) - 2\chi_1\chi_2\big]}{\sin\omega(t_2-t_1)}\ ,$$

즉

$$\boxed{S_c = \frac{m\omega}{2}\ \frac{\big[\ (\chi_1^2+\chi_2^2)\cos\omega(t_2-t_1) - 2\chi_1\chi_2\big]}{\sin\omega(t_2-t_1)}}$$